Exact-Rational Certificates in Podium: Constructions, Proofs, and Prior Art

A technical note

Adi Oltean*

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Abstract

Guidance and control software relies on numerical optimization and floating-point synthesis whose executions are difficult to verify directly. We study the problem of verifying solver *outputs*: given a certificate produced by an untrusted floating-point solver, we re-verify it in exact rational arithmetic, so that acceptance constitutes a proof that carries no rounding error. This note documents the exact-rational certificates in the Podium library (`podium.verify`) and applies the method to four properties: abort safety, via barrier certificates; optimality of supplied convex-program solutions, via Karush–Kuhn–Tucker conditions for quadratic and second-order cone programs; closed-loop stability, via Lyapunov ellipsoids from a discrete-time linear-quadratic regulator; and polynomial set containment, via sum-of-squares certificates. These certificates are standard; what we give is exact re-verification, together with their prior-art positioning. The QP and SOCP diagnostic flags may use tolerances; their certified bounds require the stated dual-feasibility conditions, with exact stationarity and exact conic dual feasibility on the SOCP path. The other certificate checks are tolerance-free. A reference implementation tests representative instances.

For nonconvex QCQPs we specialize the exact rational recovery of a semidefinite certificate, developed for polynomial optimization by Peyrl and Parrilo and by Kaltofen et al., to the S-procedure dual. This yields a tolerance-free, machine-checkable optimality-gap bracket: a rational lower bound from the dual and a rational upper bound from an exactly feasible rational point. Coincident accepted endpoints certify global optimality; exact closure requires optimal rational witnesses. We analyze the local loss in the lower bound as at most second order in multiplier error at a nonsingular interior optimum, and first order at a singular endpoint with nonzero limiting slope. Numerical proposal error and rational-rounding error both contribute to the interior bound (Theorem 2). For several constraints the S-procedure need not be tight. On a two-ball Celis–Dennis–Tapia instance [1], its bound is -3, whereas a degree-four rational sum-of-squares certificate proves $J^\star \geq \frac{-3}{2}$. An exactly feasible point gives $J^\star \leq \frac{-17520}{11881}$; these bounds certify both an upper bound on that point's suboptimality and a strictly positive gap above the S-procedure value (Theorem 4 and Appendix A).

1 Introduction

Software increasingly makes consequential decisions by solving optimization problems numerically. A spacecraft plans a fuel-optimal approach; a controller selects a maneuver that must avoid a keep-out

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zone; an autopilot commits to a command computed by a convex solver. In each case the solver returns an answer, but the answer is a floating-point number produced by an opaque computation rather than a proof that the answer is correct or optimal. Floating-point arithmetic rounds at every step, and a computation that appears to succeed can return a value that is subtly wrong. Where a mistake is costly and cannot be undone after the fact, as in spaceflight, it is desirable for the solver's answer to carry a *certificate*: a small, separate object that an independent program can verify exactly, so that correctness rests on the verifier rather than on the solver. This note constructs such certificates for guidance and control, and studies in particular bounds on the global optimum of a keep-out QCQP primitive relevant to trajectory optimization.

A concrete instance motivates the main results. Nonconvex trajectory optimization for powered landing is commonly solved by successive convexification, which linearizes the nonconvex constraints about the current iterate and solves a convex subproblem, repeating until convergence [2]. The cited convergence result concerns stationary points of a penalized problem; correspondence to a KKT point of the constrained discretized problem requires feasibility and the cited regularity assumptions. Neither stationarity nor a solver-reported cost supplies a certified bound on global suboptimality. This note supplies such a bound for a keep-out QCQP: an exact-arithmetic certificate brackets its global optimum once both a lower certificate and an exactly feasible rational upper witness have passed their checks. When the bracket closes, that witness is globally optimal. The certificate requires neither convexity nor trust in the solver; a whole-trajectory guarantee would additionally require bounds and feasibility checks for the complete trajectory problem.

The underlying idea is general. A guidance and control stack executes a large amount of untrusted numerical code: convex solvers compute commands, semidefinite programs (SDPs) synthesize safety and stability certificates, and floating-point routines emit actuator signals. Verifying the *execution* of these components is possible only in narrow settings — for active-set quadratic-programming methods on multiparametric problems, the exact worst-case iteration count can be certified offline [3] — and such analyses are specific to one solver and one problem family, leaving the floating-point behavior of the routines themselves uncovered. We instead verify the *answer*: an untrusted floating-point tool proposes a certificate, and a trusted checker re-establishes the property in exact rational arithmetic, without floating-point computation of its own. Because the check is a short, deterministic computation, it can be re-run automatically whenever the code or data change. This matters in practice: a candidate certificate produced in floating point may satisfy its defining condition only approximately, so a tolerance-based check can accept a pair (λ, t) whose value t exceeds the true optimum, whereas the exact rational test accepts a certificate only when it holds exactly.

Contributions.

1. An exact-rational verification layer for four standard guidance and control certificates, barrier abort safety, Karush–Kuhn–Tucker (KKT) optimality of supplied convex-program solutions, Lyapunov stability, and sum-of-squares set containment (Section 3), each synthesized by an untrusted solver and re-checked over the rationals $\mathbb{Q}$. Approximate QP solutions can receive an explicit rational bound under the stated hypotheses; approximate SOCP solutions receive exact residuals, with a bound only at exact stationarity and conic-dual feasibility.
2. A specialization of exact rational SDP-certificate recovery to the S-procedure dual of a nonconvex quadratically constrained quadratic program (QCQP), giving a tolerance-free machine-checkable optimality-gap bracket for guidance and control (Section 4). We demonstrate it on a keep-out primitive relevant to successive convexification (Sections 4 and 5); Section 4.5 states what a trajectory application would additionally require.

3. A multiplier-error sensitivity analysis of that bracket: the certified lower bound loses value at most quadratically in multiplier error at a nonsingular interior maximizer (Theorem 2) and at first order at a singular endpoint with nonzero limiting slope (Theorem 3), with exact acceptance in both. The analysis in Theorem 2 separates numerical proposal error from rational-rounding error. The bracket width bounds the suboptimality of the feasible point and also the duality gap (Theorem 4); a degree-four certificate proves the gap strictly positive on the two-ball example (Appendix A).
4. An open reference implementation whose trusted checker uses no floating point, with tests that reconstruct representative certificates over $\mathbb{Q}$ (Section 5).

None of the underlying certificates is new. The barrier certificates are due to Prajna and Jadbabaie [4] and Ames et al. [5]; the Lyapunov re-verification follows Wang et al. [6]; the S-lemma is due to Yakubovich [7] and its exactness is surveyed by Pólik and Terlaky [8]. Most importantly, the recovery of an exact rational certificate from a floating-point SDP solution, and its use to certify global lower bounds in polynomial optimization, is due to Peyrl and Parrilo [9], Kaltofen et al. [10], and Magron and Safey El Din [11]; Davis and Papp [12] recover exact rational certificates directly from a numerical dual certificate and prove a convergence rate, and rigorous bracketing of a relaxation value is provided by verified SDP [13, 14]. A nonconvex QCQP is the degree-two case of this machinery, and its exactness and hidden convexity are themselves classical [15, 16, 8]. Our contribution is therefore not exact rational certification, the bracket idea, nor certificate recovery with a rate, but the specialization of these to the S-procedure dual of a QCQP, in particular the local multiplier-error bounds and rational-witness conditions of Section 4, together with exact checks for a QCQP primitive relevant to guidance and control.

2 Exact re-verification of solver output

A *certificate* is a small object that makes a property easy to confirm even when the property was hard to establish: a Lyapunov matrix for stability, a multiplier for a bound, a sum-of-squares decomposition for nonnegativity. Our method separates search from proof. An untrusted floating-point routine searches for the certificate, typically by solving an SDP or an interior-point problem. The result is then rounded to nearby rational numbers and checked by an exact computation over $\mathbb{Q}$.

The single trusted primitive is an exact test for positive semidefiniteness (PSD), a symmetric $LDL^\top$ elimination over $\mathbb{Q}$ in $O(n^3)$ operations: a symmetric matrix is PSD if and only if every pivot is nonnegative and each zero pivot leaves the rest of its row, and hence by symmetry its column, in the Schur complement zero. The resulting factorization $M = LDL^\top$ with $D \succeq 0$ is itself the nonnegativity witness, since $z^\top M z = \sum_i D_{ii}(l_i^\top z)^2$. This avoids eigenvalues, which are in general irrational and admit no exact rational representation, and the leading-minor test of Sylvester, which certifies strict definiteness only. The trusted checker uses only exact rational arithmetic, so its conclusions carry no rounding error. For the barrier, Lyapunov, sum-of-squares, and optimality-gap certificates, acceptance is an exact identity together with the PSD test, with no tolerance. When the object under test is the output of an approximate solver, as for the supplied convex-program points below, the residuals are still computed exactly, but a rigorous bound on suboptimality is reported only when those exact residuals yield a valid Lagrangian dual point, not merely when they are small.

3 Four certificate families

Barrier certificates for abort safety. For linearized relative motion with a spherical keep-out zone, an SDP produces a barrier function whose sublevel set is invariant under the passive, thrust-free drift. The checker verifies over $\mathbb{Q}$ an S-procedure condition, in which a nonnegative multiplier certifies that one quadratic is nonnegative on the region where another is nonnegative. The barrier is expressed on a basis of quantities conserved by the passive drift, so its Lie derivative — the rate of change of the barrier along the drift — vanishes by construction. Together these establish forward invariance of the safe set, so that the abort set never intersects the keep-out zone [4, 5].

KKT certificates for supplied convex-program solutions. The library exposes checks for a quadratic program (QP) and a second-order cone program (SOCP). Tests exercise representative instances; the guidance planners do not invoke these checkers. Given a supplied primal-dual point, the checker computes KKT residuals, stationarity, primal and dual feasibility, and complementary slackness, exactly. A small stationarity residual is not by itself a bound on suboptimality: with a rank-deficient objective (a linear or minimum-fuel program) an approximately stationary point can be arbitrarily far from optimal. The checker therefore reports a rigorous bound only when a valid Lagrangian dual point exists, namely when the multiplier is dual-feasible, the Hessian P is PSD, and the stationarity residual r lies in the range of P. The exact objective-minus-dual-value quantity $\mu^\top(h - Gx) + \nu^\top(b - Ax) + \frac{1}{2}\, r^\top P^+ r$ then bounds $p(x) - p^\star$, the term $\frac{1}{2}\, r^\top P^+ r$ being the objective drop from the candidate to the exact minimizer of the Lagrangian, computed by solving $Pz = r$ over the rationals. If x is only feasible within a tolerance, $p(x) - p^\star$ can be negative; a feasible upper certificate requires exact primal feasibility. A SOCP objective is linear and so has no curvature to absorb a residual; a rigorous bound there instead requires exact conic-dual feasibility. Membership of a slack vector $s = (s_0, s_{1:}) \in \mathbb{R} \times \mathbb{R}^d$ in the second-order cone is decided without square roots through the algebraic condition $s_0 \geq 0$ and $\|s_{1:}\|^2 \leq s_0^2$. The exact conic predicate gates the bound; diagnostic membership flags also expose a tolerance-based check. The QP result likewise exposes a tolerance-based `certified(tol)` flag; that flag does not replace the exact conditions for a rigorous bound. The QP path verifies a PSD objective Hessian; the SOCP path has a linear objective. These conditions support global, rather than merely stationary, bounds.

Lyapunov certificates for closed-loop stability. A cost-to-go matrix P from a discrete-time linear-quadratic regulator (LQR) is re-verified as positive definite, so that its sublevel sets are bounded ellipsoids, and as satisfying the Lyapunov decrease condition $P - A_{\mathrm{cl}}^\top P A_{\mathrm{cl}} \succeq 0$, where $A_{\mathrm{cl}} = A - BK$ is the closed-loop transition matrix under the state-feedback gain K, using the same exact PSD test, in the sense of [6].

Sum-of-squares (SOS) certificates for set containment. The nonnegativity of a polynomial $p(x)$, for instance that a certified attitude cone contains a given sublevel set, is verified as an identity $p(x) = z(x)^\top G\, z(x)$ with G symmetric positive semidefinite, where $z(x)$ is a vector of monomials and G is the associated Gram matrix. Such a representation exhibits p as a sum of squares, and hence as nonnegative. The Gram matrix is recovered from an approximate SDP solution by the rational rounding of Peyrl and Parrilo [9].

4 Optimality-gap bounds for nonconvex QCQPs

Quadratically constrained quadratic programs arise throughout guidance: keep-out zones, thrust limits, and terminal ellipsoids are quadratic, and many convex subproblems inside successive convexification loops use QP or second-order-cone formulations. When the constraints are nonconvex the problem is NP-hard in general, and a solver can return a feasible point but not, on its own, a proof that no better point exists. We construct such a proof in exact arithmetic: a lower bound t and an upper bound u with $t \le J^\star \le u$, both rational and both checked without floating point, so that $u - t$ is a certified upper bound on suboptimality and the case $t = u$ certifies a global optimum (Figure 1).

Consider the QCQP

$$J^\star = \inf_{x \in \mathbb{R}^n} \Big\{ f_0(x) : f_k(x) \ge 0,\ k = 1, \dots, m \Big\}, \qquad f_j(x) = x^\top P_j x + q_j^\top x + r_j, \tag{1}$$

with symmetric $P_j \in \mathbb{Q}^{n \times n}$, $q_j \in \mathbb{Q}^n$, and $r_j \in \mathbb{Q}$ for $j = 0, \dots, m$. Submitted multipliers, lower bounds, and upper witnesses are rational; the optimization variables and dual optima need not be. A keep-out constraint $\|x - c\| \ge R$ renders the feasible set nonconvex. Throughout we use the instance $\min \|x\|^2$ subject to $\|x - (3, 0)\| \ge 5$, whose origin lies inside the excluded ball; its global optimum is $J^\star = 4$, attained at $(-2, 0)$ with multiplier $\lambda^\star = 2/5$. Table 1 lists this and the other cases considered below.

4.1 The bracket and its soundness

The lower bound is a Lagrangian relaxation, certified exactly. For multipliers $\lambda \in \mathbb{R}^m_{\ge 0}$ the Lagrangian $L(x, \lambda) = f_0(x) - \sum_k \lambda_k f_k(x)$ is quadratic in x, and with the homogenized coordinate $z = [x; 1] \in \mathbb{R}^{n+1}$ the shifted form $L(x, \lambda) - t$ equals $z^\top M(\lambda, t) z$, where

$$M(\lambda, t) = \begin{bmatrix} P_0 - \sum_k \lambda_k P_k & \frac{1}{2}(q_0 - \sum_k \lambda_k q_k) \\ \frac{1}{2}(q_0 - \sum_k \lambda_k q_k)^\top & r_0 - \sum_k \lambda_k r_k - t \end{bmatrix}. \tag{2}$$

Theorem 1 (Lower bound)**.** *If $\lambda \ge 0$ componentwise and $M(\lambda, t) \succeq 0$, then $t \le J^\star$.*

Proof. For any x, writing $z = [x; 1]$, expansion gives $z^\top M(\lambda, t) z = f_0(x) - \sum_k \lambda_k f_k(x) - t$. Positive semidefiniteness of M makes this nonnegative for all x. At a feasible x, each $f_k(x) \ge 0$ and each $\lambda_k \ge 0$, so $\sum_k \lambda_k f_k(x) \ge 0$ and hence $f_0(x) \ge t$. Taking the infimum over feasible x gives $t \le J^\star$. □

The argument is Lagrangian weak duality and uses no convexity, constraint qualification, or boundedness. Conversely, on the primal side, any point $\bar{x}$ with $f_k(\bar{x}) \ge 0$ for all k gives $J^\star \le f_0(\bar{x})$. The feasibility of $\bar{x}$ must be checked exactly: a point that is feasible only within a solver tolerance may have objective value below $J^\star$, which would place the upper bound below the true optimum. The checker therefore evaluates $f_k(\bar{x}) \ge 0$ over the rationals. By elementary gap closure, if a certified lower bound t equals a certified upper bound $u = f_0(\bar{x})$ then $J^\star = t = u$, an exact certificate of the global optimum. The two bounds are the rational S-procedure lower certificate and a rational feasible point, standard ingredients of verified global optimization; the content of this section is their specialization to the QCQP dual and the multiplier-error analysis below.

4.2 Recovery of an exact bound from an approximate dual

Fix $m = 1$ and write $A(\lambda) = P_0 - \lambda P_1$, $b(\lambda) = q_0 - \lambda q_1$, and $c(\lambda) = r_0 - \lambda r_1$. Define the extended-real dual function $g(\lambda) = \inf_{x \in \mathbb{R}^n}(f_0(x) - \lambda f_1(x))$ for all real λ; only $\lambda \ge 0$ is dual-feasible. The set

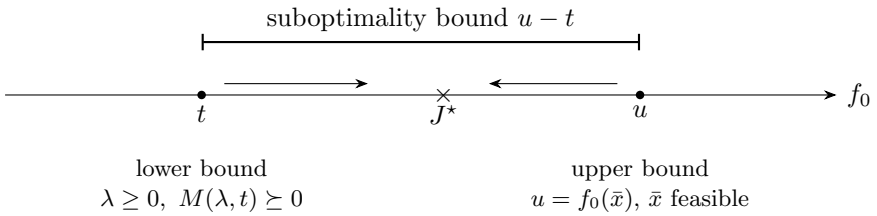


Figure 1: The bracket. A rational multiplier $\lambda \geq 0$ with $M(\lambda, t) \succeq 0$ gives a certified lower bound $t \leq J^\star$; an exactly feasible point gives a certified upper bound $u = f_0(\bar{x}) \geq J^\star$. Both are exact rationals, so $u - t$ is a certified upper bound on suboptimality; when $t = u$ the global optimum is certified exactly. The arrows indicate refinement of the dual and of the feasible point.

$U = \{\lambda : A(\lambda) \succ 0\}$ is the domain of the inverse formula below, not the full domain of g. In fact,

$$g(\lambda) = \begin{cases} c(\lambda) - \frac{1}{4} b(\lambda)^\top A(\lambda)^\dagger b(\lambda), & A(\lambda) \succeq 0,\ b(\lambda) \in \operatorname{range} A(\lambda), \\ -\infty, & \text{otherwise.} \end{cases}$$

Here $\dagger$ denotes the Moore–Penrose inverse. At a rational multiplier with finite g, one can instead solve $Aw = b$ over $\mathbb{Q}$ and evaluate $g = c - b^\top w/4$; this does not require computing a pseudoinverse. We suppress the arguments of b and c within individual formulas.

Lemma 1. *For $\lambda \in U$, $g(\lambda) = c - \frac{1}{4} b^\top A(\lambda)^{-1} b$, and $M(\lambda, t) \succeq 0$ if and only if $t \leq g(\lambda)$.*

Proof. On U the quadratic $f_0 - \lambda f_1 = x^\top A x + b^\top x + c$ is strictly convex, with unique minimizer $x(\lambda) = -A^{-1}b/2$ and minimum value $c - b^\top A^{-1} b/4$. The Schur complement of A in $M(\lambda, t)$ is $c - t - b^\top A^{-1} b/4$, so $M(\lambda, t) \succeq 0$ exactly when this scalar is nonnegative. □

On U, Cramer's rule makes g a rational function of λ with rational coefficients. Thus a rational $\lambda \in U \cap \mathbb{R}_{\geq 0}$ yields a rational lower bound that the exact PSD test certifies. For a single constraint with a strictly feasible point and finite $J^\star$, the S-lemma gives strong duality, $J^\star = \max_{\lambda \geq 0} g(\lambda) = g(\lambda^\star)$ [7].

Theorem 2 (Nonsingular recovery)**.** *Suppose $A(\lambda^\star) \succ 0$ and $\lambda^\star > 0$ maximizes the dual. Then $g(\lambda^\star) = J^\star$ and there is a neighborhood of $\lambda^\star$ in which*

$$0 \leq J^\star - g(\lambda) \leq C|\lambda - \lambda^\star|^2.$$

Rounding the exact multiplier to a rational $\widehat{\lambda}_D$ of denominator at most D with error at most $1/D$ gives an exactly certified bound with loss $O(D^{-2})$ for sufficiently large D. If instead the numerical proposal $\widetilde{\lambda}_D$ has error $\eta_D = |\widetilde{\lambda}_D - \lambda^\star|$, rounding with error at most $1/D$ gives

$$J^\star - g(\widehat{\lambda}_D) = O\big((\eta_D + D^{-1})^2\big),$$

provided the total error is small enough to remain in that neighborhood. In particular, $\eta_D = O(D^{-1})$ or exact recovery is sufficient for the rate $O(D^{-2})$; a denominator cap alone gives no such guarantee. If $\lambda^\star$ is rational, so are $J^\star$ and the corresponding optimizer, and their certificates close the bracket exactly.

Proof. On U, g is smooth and concave, with unique Lagrangian minimizer $x(\lambda) = -A(\lambda)^{-1} b(\lambda)/2$. Differentiating the minimum gives $g'(\lambda) = -f_1(x(\lambda))$. Since $\lambda^\star > 0$, $g'(\lambda^\star) = 0$, so $x(\lambda^\star)$ is feasible

with $f_1 = 0$ and $f_0(x(\lambda^\star)) = g(\lambda^\star) = J^\star$ by weak duality. Choose a closed neighborhood inside $U \cap \mathbb{R}_{\geq 0}$; boundedness of g'' and Taylor's theorem give the quadratic bound. The triangle inequality gives $|\widehat{\lambda}_D - \lambda^\star| \leq \eta_D + 1/D$. In this neighborhood Lemma 1 certifies the rounded bound exactly. If $\lambda^\star$ is rational, the rational inverse formula gives both $x(\lambda^\star)$ and $g(\lambda^\star)$ over $\mathbb{Q}$. □

Remark. With several constraints and a positive definite Lagrangian Hessian at a dual maximizer, $\partial g/\partial \lambda_k = -f_k(x(\lambda))$. For $\lambda_k^\star > 0$ this derivative is zero. At $\lambda_k^\star = 0$ it is nonpositive; it is strictly negative for a strictly slack constraint, but can be zero for a weakly active constraint. A zero multiplier does not by itself imply strict slackness.

For the running instance,

$$4 - g(\lambda) = \frac{25(\lambda - 2/5)^2}{1 - \lambda}, \qquad 0 \leq \lambda < 1.$$

The reference implementation rounds the fixed floating-point proposal $\widetilde{\lambda} = 0.4003$ to $\lambda^\star = 2/5$ at denominator cap 100, recovering $t = 4$ exactly. At cap 10000 it instead returns $\widehat{\lambda} = \frac{4003}{10000}$ with positive loss $\frac{3}{799600}$. Increasing the cap for a fixed inaccurate proposal need not improve the bound; eventually the rounded value equals that floating-point number's exact binary rational representation, leaving a nonzero error floor.

4.3 The singular case

Theorem 3 (Singular recovery)**.** *Suppose $J^\star$ is finite and a dual maximizer $\lambda^\star \geq 0$ satisfies $g(\lambda^\star) = J^\star$, with $A^\star = A(\lambda^\star) \succeq 0$ singular and $b^\star = b(\lambda^\star) \in \operatorname{range} A^\star$. For some $\sigma \in \{-1, 1\}$ and $\epsilon > 0$, suppose $\lambda^\star + \sigma\delta \in U \cap \mathbb{R}_{\geq 0}$ for $0 < \delta < \epsilon$, the finite limit*

$$\bar{x} = \lim_{\delta \downarrow 0} -\tfrac{1}{2} A(\lambda^\star + \sigma\delta)^{-1} b(\lambda^\star + \sigma\delta)$$

exists, and $f_1(\bar{x}) \neq 0$. Then: (a) a rational $\lambda^\star$ yields a rational exact lower certificate $(\lambda^\star, J^\star)$, accepted even though its matrix is singular; (b) $\sigma f_1(\bar{x}) > 0$ and

$$J^\star - g(\lambda^\star + \sigma\delta) = |f_1(\bar{x})|\, \delta + o(\delta),$$

so admissible rational multipliers approaching the endpoint yield bounds whose loss is first-order in multiplier error; (c) exact rational bracket closure additionally requires a rational feasible optimizer, and is impossible if $J^\star$ is irrational.

Proof. (a) At a rational endpoint solve $A^\star w = b^\star$ over $\mathbb{Q}$. Then $J^\star = c(\lambda^\star) - (b^\star)^\top w/4$ is rational and

$$L(x, \lambda^\star) - J^\star = (x + w/2)^\top A^\star (x + w/2).$$

This identity gives $M(\lambda^\star, J^\star) \succeq 0$. The exact $LDL^\top$ test accepts a zero pivot precisely when the remaining pivot row vanishes, and thus accepts singular PSD matrices. (b) Taking limits in the normal equations gives $A^\star \bar{x} = -b^\star/2$, so $\bar{x}$ minimizes the endpoint Lagrangian. Consequently $g(\lambda^\star + \sigma\delta) \to g(\lambda^\star) = J^\star$. Along the positive definite branch the derivative with respect to δ is $-\sigma f_1(x(\lambda^\star + \sigma\delta))$, which tends to $-\sigma f_1(\bar{x})$. A first-order expansion at the endpoint proves the displayed loss formula; maximality and the nonzero hypothesis force $\sigma f_1(\bar{x}) > 0$. (c) A closed rational bracket supplies a rational feasible optimizer with objective equal to its rational lower endpoint, hence a rational $J^\star$. Conversely, an optimal rational lower certificate and a rational feasible optimizer close the bracket. □

Remark (The nullspace component). Let $N = \ker A^\star$ be invariant under P_1. The adjacent positive definite interval makes $P_1|_N$ invertible, and the normal equations on N and $N^\perp$ give

$$\bar{x} = -\tfrac{1}{2}(A^\star)^\dagger b^\star - \tfrac{1}{2}(P_1|_N)^{-1}\Pi_N q_1. \tag{3}$$

Indeed, $\Pi_N b^\star = 0$, while the N blocks of A and b are $-\sigma\delta P_1|_N$ and $-\sigma\delta\Pi_N q_1$. The second term vanishes if $\Pi_N q_1 = 0$, as in the centered trust-region case $P_1 = -I$, $q_1 = 0$; $P_1 = -I$ alone does not suffice. For a centered ball, $f_1(\bar{x}) > 0$ describes the classical hard case [17].

The theorem describes local sensitivity to multiplier error and preserves the soundness of supplied singular certificates. The implementation's instance $\min -x_1^2 + 2x_2^2 + 2x_2$ subject to $1 - \|x\|^2 \geq 0$ has $A(\lambda) = \mathrm{diag}(-1+\lambda, 2+\lambda)$, singular at $\lambda^\star = 1$, with $J^\star = -4/3$. For $\lambda = 1 + \delta$, $J^\star - g(\lambda) = \delta[1 - 1/(3(3+\delta))]$, with limiting slope $8/9$. The exact certificate $(1, -4/3)$ is accepted, but the implementation's positive-definite-only recovery routine does not evaluate that endpoint. Its primal optimizers $(\pm 2\sqrt{2}/3, -1/3)$ are irrational, so a rational upper witness cannot close this bracket even though its optimal lower certificate is rational. Applying the change of variable $x_1 \mapsto x_1 - a$ to both objective and constraint, with rational a, gives $\bar{x} = (a, -1/3)$, whereas the pseudoinverse term alone gives $(0, -1/3)$, illustrating the extra term in (3).

4.4 Several constraints

Define the S-procedure dual value

$$d^\star = \sup\{t : \lambda \in \mathbb{R}^m_{\geq 0},\ t \in \mathbb{R},\ M(\lambda, t) \succeq 0\}.$$

It is distinct from the Shor *primal* relaxation value v_{Shor}, obtained by replacing $xx^\top$ by X and imposing $\left[\begin{smallmatrix} X & x \\ x^\top & 1 \end{smallmatrix}\right] \succeq 0$. SDP weak duality gives $d^\star \leq v_{\mathrm{Shor}}$, and relaxation gives $v_{\mathrm{Shor}} \leq J^\star$; identifying the first two values requires SDP strong duality, for example a suitable Slater condition with finite relaxation value. We do not impose such regularity in the next result.

Theorem 4 (Certified bounds on suboptimality and duality gap)**.** *For every m, a rational certificate $\lambda \geq 0$, $M(\lambda, t) \succeq 0$ and a rational exactly feasible point $\bar{x}$, with $u = f_0(\bar{x})$, give*

$$t \leq d^\star \leq J^\star \leq u.$$

In particular,

$$0 \leq u - J^\star \leq u - t, \qquad 0 \leq J^\star - d^\star \leq u - t.$$

Thus $[t, u]$ encloses the optimum, and its width bounds both the suboptimality of $\bar{x}$ and the S-procedure duality gap. If $t = u$, these certificates establish global optimality. For $m \geq 2$, $d^\star < J^\star$ is possible.

Proof. The supplied multiplier and bound are feasible for the supremum defining $d^\star$, so $t \leq d^\star$. Theorem 1 holds for every nonnegative real multiplier; taking the supremum gives $d^\star \leq J^\star$. Exact feasibility gives $J^\star \leq u$. These finite endpoints imply both displayed gap bounds by subtraction. If the endpoints coincide, all intermediate quantities coincide. The two-ball example below proves that strict S-procedure gaps can occur with two constraints. □

Remark. For $m = 1$ with a strictly feasible point and finite $J^\star$, the S-lemma gives $d^\star = J^\star$. This equality over $\mathbb{R}$ does not imply that optimal lower certificates or feasible optimizers exist over $\mathbb{Q}$. For instance, $\min(x-2)^2$ subject to $2 - x^2 \geq 0$ has irrational optimum $6 - 4\sqrt{2}$, so no rational bracket closes exactly despite a nonsingular dual optimum. For several constraints, the optimal *real* SOS bounds in the moment/SOS hierarchy converge to $J^\star$ under an Archimedean assumption [18].

Turning a numerical candidate into an exact rational lower certificate additionally requires rational Gram matrices, exact coefficient identities, and PSD verification. Real convergence alone does not guarantee that arbitrary rounding produces such certificates.

For the two-ball Celis–Dennis–Tapia instance [1]

$$\min 2x_1x_2 \quad \text{subject to} \quad \|x-(1,0)\| \le 2, \quad \|x+(1,0)\| \le 2,$$

the S-procedure dual has value $d^\star = -3$. Its rational upper witness gives the exact enclosure $[-3, \frac{-17520}{11881}]$. Appendix A supplies a degree-four rational SOS identity proving $J^\star \ge \frac{-3}{2}$. Hence

$$J^\star \in [\frac{-3}{2}, \frac{-17520}{11881}], \qquad 0 \le u - J^\star \le \frac{603}{23762},$$

and

$$\frac{3}{2} \le J^\star - d^\star \le \frac{18123}{11881}.$$

The positive lower endpoint proves a strict S-procedure duality gap exactly. The higher-degree certificate improves the objective enclosure without changing $d^\star$, which refers only to the original quadratic relaxation.

On the two-ball instance of Appendix A, the Shor primal attains -3 at $x = 0$ and $X = \frac{3}{2}\left[\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right]$. Thus the positive certified QCQP relaxation gap there coexists with zero SDP primal–dual gap. In the note's revision artifacts, `revision-2026-09-08/verify_revision.py` computes this Shor witness's PSD verdict, constraint slacks and objective. It assembles the dual minorant matrix from the quadratic data and supplied multipliers, checks its PSD verdict and complementary inner product, and replays the exact degree-four witness check. Its singular-endpoint controls retain exact minimizer sequences and derive limiting points from both diagonal and rationally rotated problem data. These checks are separate from the library tests. The earlier `revision-2026-09-06/run_checks.py` invokes the manuscript gates from the paper repository and runs library tests; those gates are outside the revision artifact folder. That runner does not check the Shor witness.

4.5 Relation to successive convexification

Nonconvex trajectory optimization in guidance is commonly addressed by successive convexification and sequential convex programming, which repeatedly convexify about the current iterate and solve convex subproblems [2]. Their convergence statements depend on the method's assumptions: a stationary point need not be locally optimal, and feasibility of the original trajectory problem must be established. A returned iterate is therefore a proposal for an upper witness. To use it here, a rational point $\bar{x}$ must satisfy every constraint of the original rational QCQP exactly; rational rounding can destroy feasibility and must be checked.

For that same QCQP, an accepted S-procedure certificate gives a lower bound $t \le J^\star$, so $u - t$ bounds the global suboptimality of the accepted point. A trajectory-wide claim requires an objective and constraints describing the entire trajectory problem, including a justified treatment of any nonquadratic or continuous-time conditions. The demonstrated application here is the finite-dimensional keep-out QCQP primitive. Certification of a convex subproblem or of that primitive alone does not establish a global bound for an entire nonconvex trajectory optimization problem.

5 Reference implementation

The reference library provides the exact rational arithmetic and PSD checks used by these certificates. Library tests exercise the QCQP examples; the separate revision checks verify the degree-four CDT

Table 1: Representative instances. All certified bounds are exact rationals verified over $\mathbb{Q}$; "closed" indicates that the certified lower bound t and upper bound u coincide and hence certify $J^\star$. The value $\widetilde{\lambda} = 0.4003$ is the approximate dual fed to the rational rounding step; the recovered certificate is exact. The CDT row reports a degree-four Gram certificate, so its multiplier column is inapplicable; the quadratic S-procedure optimum still has $\lambda = (1/2, 1/2)$ and value -3.

Instance	Case	$\lambda^\star$	certified $[t, u]$	$J^\star$
One keep-out	nonsingular, closed	$2/5$	$[4, 4]$	4
Same, non-optimal upper point	open bracket	$2/5$	$[4, 169/25]$	4
Recovery from $\widetilde{\lambda} = 0.4003$	exact at $D = 100$	$2/5$	$t = 4$	4
Centered hard case	singular	1	$t = -4/3$	$-4/3$
Two active keep-outs	several constraints	$(\frac{1}{2}, \frac{1}{2})$	$[5, 9]$	5
Two-ball CDT	degree-four SOS bound	—	$[\frac{-3}{2}, \frac{-17520}{11881}]$	≈ -1.476

certificate. Table 1 collects representative cases from both. These finite exact checks do not establish the general analytic theorems. The two-active-keep-out row minimizes $\|x\|^2$ outside both radius-3 balls centered at $(2, 0)$ and $(-2, 0)$. Two requirements that a tolerance-based check would miss are enforced exactly. First, the checker verifies feasibility of $\bar{x}$ over the rationals, since a point feasible only within a tolerance may have objective value below $J^\star$. Second, the closing criterion draws its two bounds from a single problem instance, since bounds computed for different instances cannot be compared.

Mechanization. The pointwise fixed-multiplier inequalities $t \le f_0(x)$ for feasible x used by Theorems 1 and 4, the latter's pointwise suboptimality inequality, and the scalar case of Lemma 1 have machine-checked counterparts in a Lean 4 corpus (`note_thm_sound`, `note_thm_multi`, `note_bracket`, `note_lem_schur`). The companion note [19] maps each declaration to its result here and records the scope limits: the mechanization takes the PSD-to-minorant lifting step as a hypothesis rather than proving it. The infimum step yielding $t \le J^\star$ remains on paper; the supremum $d^\star$, the inequality $d^\star \le J^\star$, the strict-gap example and the note's remaining results are not mechanized.

6 Related work

Exact and certified certificates for global polynomial optimization are an established subject. The moment/SOS hierarchy of Lasserre [18] and Parrilo [20] gives convergent real lower bounds under its regularity assumptions, Henrion and Lasserre [21] extract optimizers and detect global optimality, and Nie, Demmel, and Sturmfels [22] give SOS exactness over the gradient ideal. Exact rational certificates recovered from a numerical SDP are due to Peyrl and Parrilo [9], Kaltofen et al. [10], and Magron and Safey El Din [11]; closest to our rate statement, Davis and Papp [12] construct exact rational certificates from numerical dual certificates and prove a convergence rate. Rigorous, though not exact, bracketing of the relaxation value by verified SDP is due to Jansson [13] and Roux et al. [14], and an exact SDP duality theory to Ramana [23]. A nonconvex QCQP is the degree-two instance of this machinery; its duality, exactness, and hidden convexity are classical [7, 24, 8, 15, 16, 25], as is the trust-region hard case [17, 26].

Against this background our contribution is narrow. We instantiate the exact rational recovery of Peyrl and Parrilo and of Davis and Papp on the S-procedure dual matrix $M(\lambda, t)$ of a QCQP; we

record the local sensitivity of the certified lower bound to multiplier error, at most second-order at a nonsingular interior maximizer (Theorem 2), and first-order at a singular endpoint with an adjacent positive-definite branch and nonzero limiting slope (Theorem 3). This is the standard second-order-versus-first-order local behavior at stationary interior and nonstationary boundary maximizers, stated for the QCQP dual; we do not claim it as new. The weak-duality bracket and its closure are likewise standard. The contribution is the integration: an exact-rational implementation of the QCQP primitive and its certificate checks, with the conditions for a guidance application stated in Section 4.5. Barrier and control-barrier certificates [4, 5], credible autocoding [6], and the verification of convex-optimization solvers for model-predictive control [27] are complementary in verifying control software: the last verifies the solver algorithm, whereas we verify its output. Lossless convexification [28] and successive convexification [2] address the feasibility and tightness of convex reformulations; applying the bounds here to a resulting trajectory would additionally require certificates for the complete original problem (Section 4.5).

7 Limitations and open problems

Under the hypotheses of Theorem 3, at a rational singular multiplier, the dual can be evaluated through a consistent rational solve and PSD verification. The reference recovery routine implements neither that endpoint evaluation nor the search for a singular boundary certificate from approximate data. It uses a positive-definite solve; a supplied singular pair can still pass the PSD verifier. Approaching from the inadmissible side yields no certificate. Even with an optimal lower certificate, rational bracket closure can be impossible because the optimizer or optimum is irrational. The local sensitivity results do not remove numerical proposal error or guarantee convergence as a denominator cap increases.

The degree-four certificate in Appendix A resolves strictness of the gap for that instance; it does not compute the exact irrational optimum by rational gap closure or provide a general finite-level recovery method. The implementation evidence concerns individual primitives, not an entire trajectory optimization problem. The partial Lean coverage is stated in Section 5.

The $LDL^\top$ PSD test takes $O(n^3)$ rational arithmetic operations; this is not a bit-complexity bound. Intermediate numerators and denominators can grow under exact elimination. The examples establish feasibility at their stated sizes, not scalability to arbitrary guidance problems.

8 Conclusion

We have described exact rational verification of certificates proposed by untrusted guidance and control computations, with checkers for barrier safety, KKT conditions and qualified optimality bounds, Lyapunov stability, and SOS nonnegativity. For a nonconvex QCQP, an accepted lower certificate and an exactly feasible rational upper witness enclose the global optimum. The local loss of the lower bound is at most quadratic in multiplier error at a nonsingular interior maximizer and linear at a singular endpoint with nonzero limiting slope. Approximate proposals contribute their own error. A degree-four rational SOS certificate tightens the two-ball example's enclosure and proves its S-procedure duality gap positive. These constructions are realized in an open reference implementation and accompanying exact witnesses.

A An exact certificate for the two-ball instance

Write the coordinates as (x, y) and the constraints as $g_\pm(x, y) = 3 - x^2 - y^2 \pm 2x \geq 0$. For nonnegative constant multipliers λ_+, λ_-, set $s = \lambda_+ + \lambda_-$ and $h = \lambda_- - \lambda_+$. The Lagrangian quadratic coefficient matrix is $\left[\begin{smallmatrix} s & 1 \\ 1 & s \end{smallmatrix}\right]$, so finite dual values require $s \geq 1$. For $s > 1$ the dual value is

$$-3s - \frac{h^2 s}{s^2 - 1} \leq -3s < -3.$$

At $s = 1$, the range condition forces $h = 0$, yielding $g = -3$. Thus $(\lambda_+, \lambda_-) = (1/2, 1/2)$ certifies the optimal S-procedure lower bound $d^\star = -3$.

Let $v_2 = (1, x, y, x^2, xy, y^2)^\top$ and $v_1 = (1, x, y)^\top$. The rational matrices below satisfy

$$\tfrac{3}{2} + 2xy = v_2^\top G_0 v_2 + g_+ v_1^\top G_+ v_1 + g_- v_1^\top G_- v_1. \tag{4}$$

$$G_0 = \frac{1}{500}\begin{pmatrix} 264 & 0 & 0 & -91 & 69 & -146 \\ 0 & 116 & 71 & 0 & 0 & 0 \\ 0 & 71 & 46 & 0 & 0 & 0 \\ -91 & 0 & 0 & 148 & 44 & 47 \\ 69 & 0 & 0 & 44 & 190 & 44 \\ -146 & 0 & 0 & 47 & 44 & 136 \end{pmatrix}$$

$$G_+ = \frac{1}{500}\begin{pmatrix} 81 & -27 & 57 \\ -27 & 74 & 22 \\ 57 & 22 & 68 \end{pmatrix}, \qquad G_- = DG_+D, \quad D = \mathrm{diag}(1, -1, -1).$$

Their exact $LDL^\top$ factorizations have positive pivots. Coefficient comparison verifies (4) over $\mathbb{Q}$, so every term on the right is nonnegative on the intersection of the two balls and $J^\star \geq \frac{-3}{2}$. The feasible rational point

$$\bar{x} = \left(\frac{73}{109}, \frac{-120}{109}\right), \qquad (g_+(\bar{x}), g_-(\bar{x})) = (\frac{292}{109}, 0)$$

has objective $u = \frac{-17520}{11881}$. It gives the improved objective enclosure and the duality-gap bounds stated in Section 4.

For comparison, the true optimum can also be determined algebraically. For $a = |x| \in [0, 1]$, feasible y satisfies $y^2 \leq 3 - 2a - a^2$; minimizing $2xy$ reduces to maximizing $a^2(3 - 2a - a^2)$. Its interior maximizing point solves $3 - 3a - 2a^2 = 0$, giving $a = (\sqrt{33} - 3)/4$ and

$$J^\star = -\sqrt{\frac{207 - 33\sqrt{33}}{8}}.$$

This irrational value cannot be an endpoint of a closed rational bracket. The rational identity above proves a useful bound without representing the optimizer or the optimum algebraically.

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